SHREEJA SINGH

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Education

Illinois Institute of Technology **Aug 2025 - May 2027**
Master of Science, Computer Science (GPA: 3.6/4.0) *Chicago, IL*

Bangalore Institute of Technology **Nov 2020 - Jun 2024**
Bachelor of Technology, Electronics and Communication Engineering (GPA: 4.0/4.0) *Bengaluru, India*

Work Experience

Aion Labs | *AI Engineer Intern* **May 2026 - Present**

- Designed and secured an end-to-end AWS environment for processing sensitive customer PII across 4 source systems: provisioned a VPC, network firewall with egress allow-listing, NAT/private routing, and a locked-down EC2 workspace (Session Manager, scoped IAM, no public ingress); identified and fixed a PII deny-policy gap and an AWS reserved-route conflict before launch.
- Integrated an AI coding agent (Claude Code on Amazon Bedrock) behind layered IAM controls; generated schema-matched synthetic fixtures and validated via DuckDB profiling to confirm zero access to restricted PII.
- Built and deployed 3 production AI agents on Azure VM with a RAG pipeline (ChromaDB + sentence-transformers), applying data engineering best practices and Obsidian memory vaults; configured 6 AGENTS.md files to enable cross-agent sharing and autonomous memory updates.
- Developed CI/CD pipelines using GitHub Actions and Docker to automate testing and deployment of AI agents across environments.

Microchip Technology | *Software Quality Engineer* **Jul 2024 - Jul 2025**

- Cut manual search time by 40% for the engineering team by building an AI-powered chatbot in Python (integrated with ASPICE, JIRA, POLARION, CAD) with image-recognition and NLP for process-diagram analysis.
- Drove delivery in Agile/Scrum; documented in Confluence and presented to senior leadership.

Projects

Personal AI Agent System (Aairy) **Mar 2026 - Present**

- Built a locally hosted autonomous AI assistant using Python and the Telegram bot framework; engineered a 4-layer memory architecture for cross-session persistence, informed by research design principles for data management and reproducibility.
- Implemented a RAG pipeline from scratch with ChromaDB and sentence-transformers for semantic search across all memory files, plus a nightly auto wiki-builder that extracts key facts via the Claude API and updates the vault 100% autonomously.
- Cut API costs by 89% via intelligent model routing (Claude Haiku vs. Sonnet) and context optimization, making continuous autonomous operation affordable.

Autonomous Multi-Agent AI System **Jan 2026 - Mar 2026**

- Architected an 8-agent pipeline in Python with LLM-based intent detection, routing queries across four specialist paths; migrated memory to a ChromaDB vector store for robust data assimilation and cosine-similarity retrieval; deployed via GitHub Actions CI/CD for repeatable experiments.
- Implemented an RLHF feedback loop in PyTorch, collecting 1–5 star signals to iteratively refine model performance; benchmarked quality (8.0/10) and latency (4.72s) across five query types as part of rapid experiment cycles.

Technical Skills

- **Machine Learning & Research:** Machine Learning, Scientific Machine Learning, Research Design, PyTorch-, Time-Series Forecasting, Data Assimilation, Computer Vision
- **Languages & Frameworks:** Python, FastAPI, RESTful APIs, Java
- **AI & Agentic Systems:** LLMs (Claude API, GPT, LLaMA 3.3/4 Scout, Groq), Multi-Agent Orchestration, RAG Pipelines, ChromaDB, Sentence-Transformers, RLHF, LLM-as-Judge, Prompt Engineering, OpenClaw, Agent Memory Architecture
- **Cloud, DevOps & Data:** AWS (VPC, IAM, EC2, Bedrock), Docker, Azure VM, GitHub Actions, CI/CD, PostgreSQL, MySQL, Git, API Security, Data Engineering, Geospatial Data, Weather Data, Climate Data